School of Engineering

Cochin University of Science and Technology

Division of Computer Science and Engineering

M.Tech Degree in Computer Science and Engineering (Cyber Security)

First Internal Examinations, October 2025

22-454-0101 Mathematical Foundation for Cyber Security

TIME: 2 hours **MAX. MARKS: 50**

COURSE OUTCOMES
After completion of course, students will be able to:
1. Able to apply Number theory in data security.
2. Understand the ideas of group, ring and an integral domain and be aware of examples of these structures in mathematics.
3. Understand probability and statical methods.
4. Understand coding theory and its use in data security.

Answer all questions

1. Give the pseudocode for the Euclidean algorithm to find the greatest common divisor (GCD) of two positive integers, and explain its complexity. (5)

2. Express the $GCD(12378, 3054)$as $12378x + 3054y$ where x and y are integers. (5)

3. Find $7^{222} mod\ 11$. (5)

4. What are the solutions of the linear congruence $3x \equiv 4\ (mod\ 7)$? (5)

5. Let X be a number on a die when thrown. Obtain the distribution function. Find $P(X \leq 4)$ and $P(X > 3)$. (5)

6. Suppose that $P(A|B) = 0.4$ and $P(B) = 0.5$. Determine $P(A \cap B)$ and $P(A \cap B)$. (5)

7. The current in a circuit as measured by an ammeter is a continuous random variable X with the following density function: (10)

$$f(x) = \begin{cases} 0.075x + 0.2, 3 \leq x \leq 5 \\ 0, otherwise \end{cases}$$

Verify that the total area under the density curve is 1.

Calculate $P(X \leq 4)$ and $P(3.5 \leq X \leq 4.5)$.

8. A certain number of objects is unknown. When divided by 3, the remainder is 2; when divided by 5, the remainder is 3; and when divided by 7, the remainder is 2. Using Chinese Remainder Theorem, determine the number of objects. (10)